

Braidyn Sheffield

Denver, Colorado • (702) 333-5343 • sheffieldbraidyn@gmail.com • <https://braidyn-sheffield.github.io/>

EDUCATION

University of Denver, Denver, CO

Expected: June 2026

Major / Minor: BS in Computer Engineering / Mathematics

GPA: 3.87 / 4.00

Accomplishments: Dean's List (Spring 2023 - Winter 2026)

SKILLS

TECHNICAL SKILLS:

Embedded Systems & Hardware: Microcontrollers, Arduino, Teensy, STM32, Embedded C, Real-Time Systems, State Machine Design, Sensor Integration, Motor Control, Servo Control, Ultrasonic Sensors, Color Sensors, IMU/Gyroscope Integration, Hardware Debugging, Hardware Validation, System Testing

Electrical & PCB Design:

PCB Design, Circuit Analysis, Analog Electronics, Digital Electronics, Power Systems, Voltage Regulation, Buck Converters, Wiring & Harnessing, Soldering, Signal Conditioning, Electrical Troubleshooting, Hardware Integration, Test Equipment, Multimeters, Oscilloscopes

Communication Protocols:

UART, SPI, I2C, Serial Communication, Bluetooth Communication, NFC Pairing

Software & Programming:

C, C++, Embedded C, Python, MATLAB, Verilog, Git/GitHub, Flask, TensorFlow, Google Colab, VS Code

CAD, Fabrication & Prototyping:

SOLIDWORKS, Autodesk Inventor, CAD Modeling, 3D Printing, Rapid Prototyping, Fixture Design, Electronics Prototyping, Assembly & Integration

Engineering Tools:

LabVIEW, SPICE/PSPICE, Quartus, MindView, Microsoft Office Suite, Arduino IDE, STM32CubeIDE

Project & Leadership Skills:

Project Coordination, Project Planning, Technical Documentation, Requirements Tracking, Verification & Validation, System Integration, Risk Assessment, Team Leadership, Workshop Facilitation, Cross-Functional Collaboration, Technical Presentations, Troubleshooting, Root Cause Analysis

PROJECTS

Bluetooth Infusion Pump – STM32, FreeRTOS

2025 – Present

- Designed and developed a Bluetooth-enabled infusion pump as part of an industry-sponsored capstone project with B. Braun.
- Coordinated project schedules, milestones, documentation, and subsystem integration efforts using MindView.
- Implemented a multitasking embedded system using FreeRTOS on STM32 to support real-time monitoring and control.
- Developed a touchscreen GUI on a 7-inch TFT display to visualize flow rate, temperature, and heart rate data.
- Integrated wireless communication, sensors, motor control, and power distribution subsystems into a single medical device platform.
- Performed system integration, verification, validation, and troubleshooting across hardware and software components.
- Collaborated with a multidisciplinary engineering team through design reviews, testing, and final system demonstrations.

Autonomous Delivery and Identification Vehicle – C++, Embedded Systems

2025

- Led a team of five students through the design, development, and testing of an autonomous robotic vehicle.

- Developed a modular C++ state-machine architecture to coordinate navigation, sensing, object identification, and delivery operations.
- Integrated four color sensors, three ultrasonic sensors, an MPU-6050 IMU, servos, and motor control systems.
- Designed a power distribution architecture using dual buck converters to support multiple voltage domains.
- Implemented obstacle avoidance, wall-following, line-following, and autonomous delivery behaviors.
- Conducted iterative integration testing and troubleshooting to improve reliability and navigation accuracy.
- Successfully met all competition requirements through fully autonomous operation.

Self-Balancing Robot – PCB Design, Embedded Systems

2025

- Designed and fabricated a custom PCB to replace a development board, integrating microcontroller, power regulation, and sensor interfaces into a compact system.
- Developed embedded control logic to stabilize a two-wheeled self-balancing robot using real-time sensor feedback.
- Integrated IMU sensor data to estimate tilt and orientation, enabling continuous balance correction.
- Implemented motor control algorithms to dynamically adjust wheel speed and maintain stability.
- Debugged hardware and control issues through iterative testing, improving system reliability and performance.

Simulated Garbage Collector in C – C, Memory Management

2025

- Designed and implemented a custom memory allocator in C (duMalloc/duFree), managing heap memory allocation and deallocation without relying on standard library functions.
- Developed a managed pointer system to track allocated memory and prevent leaks, improving memory safety and control.
- Implemented multi-generation garbage collection, including minor and major collection cycles, to efficiently reclaim unused memory.
- Built dynamic memory management structures using pointers and heap allocation, reinforcing low-level memory management concepts.
- Debugged segmentation faults and memory errors, validating correctness through extensive testing and iterative refinement.

EKG Circuit Design – Analog Circuits, Signal Processing

2024

- Designed a low-cost analog EKG circuit to capture heart electrical signals as part of a 2-person team addressing real-time biomedical sensing challenges.
- Researched and selected amplifier, filter, and electrode configurations, improving signal clarity and reducing electrical noise by 35%.
- Built and tested the circuit on a breadboard, comparing output waveforms (P, QRS, T waves) to medical-grade references with <5% signal distortion.
- Documented system design, testing, and medical applications in a technical report, earning top evaluation marks from faculty.

Digital Lock System – Verilog, FPGA (Quartus)

2024

- Designed and implemented a digital lock system using Verilog on an FPGA, incorporating finite state machine (FSM) logic
- Developed sequential logic to validate input sequences and control lock/unlock behavior
- Simulated and verified functionality using testbenches to ensure correct state transitions
- Integrated hardware inputs/outputs (switches, LEDs, displays) for real-time user interaction
- Debugged timing and logic errors to ensure reliable system operation

Robotic Arm – Arduino, CAD, Embedded Systems

2024

- Designed and built a multi-axis robotic arm using servos and potentiometers for real-time manual control.
- Developed embedded control logic to map potentiometer inputs to precise servo positioning.
- Modeled and fabricated mechanical components using CAD and 3D printing.
- Integrated hardware and software systems, enabling smooth and responsive motion control.
- Debugged mechanical alignment and signal noise issues to improve movement accuracy.

Digital Thermometer – Embedded Systems, Sensors, Microcontrollers

2024

- Designed and implemented a digital temperature monitoring system using embedded hardware and sensor interfaces.
- Developed software to acquire, process, and display temperature measurements in real time.
- Integrated sensing, signal conditioning, and user-interface components into a functional prototype.
- Validated measurement accuracy through testing and calibration procedures.
- Troubleshoot hardware and software integration issues to improve system reliability.

PROFESSIONAL EXPERIENCE

Entrepreneurship Garage, Denver CO – Managing Technical Assistant

January 2023 - Present

- Supervise and train 5 Technical Assistants, ensuring safe equipment use and efficient operation across three engineering makerspaces.
- Coordinate and lead 10+ workshops per quarter covering Arduino, electronics, CAD, 3D printing, and rapid prototyping, training more than 200 students annually.
- Assist students in designing, prototyping, troubleshooting, and validating 40+ engineering projects involving embedded systems, electronics, CAD, and fabrication.
- Support project planning, material selection, and manufacturing decisions throughout the design process.
- Operate and maintain fabrication equipment including 3D printers, soldering stations, electronics workbenches, and rapid prototyping tools.
- Provide technical mentorship on circuit design, microcontrollers, PCB design, and hardware integration.
- Coordinate equipment scheduling, resource allocation, and makerspace organization to support multiple concurrent engineering projects.

Atomic Motors, Las Vegas NV – Automotive Technician

June 2021 – June 2022

- Diagnosed and repaired electrical systems on classic vehicles through schematic interpretation and systematic troubleshooting.
- Installed wiring harnesses, electronic fuel injection systems, air conditioning systems, gauges, and auxiliary electrical components.
- Utilized multimeters and diagnostic equipment to identify electrical faults and verify repairs.
- Performed mechanical and electrical integration of aftermarket systems while ensuring reliability and functionality.

US Navy, Atsugi Japan – Aviation Electronics Technician

July 2017 – May 2021

- Troubleshoot and repaired radar, navigation, communication, and mission systems on military aircraft, completing over 3,000 maintenance actions.
- Supported 550+ mishap-free flight hours by ensuring aircraft systems remained fully operational and mission ready.
- Performed system testing, calibration, fault isolation, and verification using technical manuals and electronic test equipment.
- Conducted pre-flight and post-flight inspections to evaluate aircraft readiness and identify system deficiencies.
- Led and trained junior technicians in avionics troubleshooting, maintenance procedures, safety protocols, and technical documentation.
- Maintained maintenance records and technical documentation in accordance with military aviation standards.
- Coordinated maintenance activities with flight crews, maintenance control, and engineering personnel to meet operational requirements.
- Improved troubleshooting and tool-control processes, reducing downtime and increasing inspection readiness.

Leadership & Involvement

Vice President – University of Denver Math Club

- Lead club operations, event planning, and member engagement activities.
- Coordinate academic and social events supporting student success and professional development.

Student Coordinator – University of Denver Equity in STEM

- Assisted in planning and coordinating student engagement activities, workshops, and program initiatives to support engineering student success.

Member – University of Denver Tau Beta Pi Engineering Honor Society

- Inducted into the nation's oldest engineering honor society in recognition of academic achievement, engineering excellence, and exemplary character.

Member – University of Denver Society of Hispanic Professional Engineers (SHPE)

- Active member participating in professional development, networking, and engineering outreach events.

- Attended regional and national SHPE conferences to engage with industry professionals and explore engineering career opportunities.

Peer Leader – University of Denver Equity in STEM

- Guided incoming engineering students through their transition to university life by providing academic support, mentorship, and campus resource guidance.